Chapter 6

Fine-Tuning Content with Localization and Translation

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No matter what kind of content creation you do, if you want to reach a global audience, you must keep localization and translation in mind. *Translation*, or the process of translating words from one language to another, is just a small part of the larger process known as localization. *Localization* refers to the process of making a product, article, video, or anything involving language meet the needs of a particular audience. In addition to translation, localization also takes into account the culture, expectations, local standards, and legal requirements of the target market.

Businesses that market in multiple countries can't run effectively without strong localization and translation strategies. Historically, businesses applied these efforts to localizing and translating the company's website content, but they now extend the practice to all digital advertising and many customer relationship management (CRM) programs, too. AI can play an important role in making the process of localization and translation more efficient and effective.

Consider a multinational e-commerce platform, such as Amazon, which localizes its website for different countries so that it can enhance the shopping experience worldwide. Amazon accomplishes this goal by translating product descriptions, adapting currency and measurement units, and tailoring the user interface to reflect local tastes and legal requirements. Customers expect this localization from every multinational brand from which they purchase. This level of localization not only makes a company accessible but also builds trust and loyalty among users by showing respect for their language and culture.

Leveraging localization and translation activities to adapt your business content ensures that digital advertising, marketing campaigns, products and services, and all communications are relevant and user-friendly across various geographical and linguistic landscapes. In this chapter, I describe underlying principles of cultural awareness, strategies, and best practices for using AI in your localization and translation efforts, and the tools that you can find to help you along the way.

Exploiting AI for Localization and Translation

Using AI in your approach to localization efforts can make those efforts more efficient and workable, whether you're a marketer for a small business or a Fortune 500 company. At a basic level, AI-powered translation services, such as Google Translate, can drastically improve the speed and accuracy of translating web content on-the-fly. These services enable small businesses to reach international markets without the prohibitive costs of traditional translation services.

AI can also tailor digital advertising and marketing based on geographic and linguistic data, allowing companies to run localized campaigns that resonate with the target audience. For instance, Netflix uses AI not only to translate but also to recommend content by analyzing viewing patterns in different regions. This two-pronged approach helps to ensure that its suggestions are culturally relevant and engaging for each unique user base.

Capturing cultural context

Understanding the cultural context of your intended audience is crucial when localizing and translating so that you can create effective digital content — your website copy, social media campaigns, or CRM newsletters, for example. You must ensure that the adapted content not only translates linguistically, but also aligns with local customs, values, and behaviors. Infusing your translations with an attention to culture can significantly impact audience reception and engagement.

For instance, a social media campaign that acknowledges local holidays and traditions can foster a stronger connection with the audience than generic content developed by a marketing team and then translated. Similarly, CRM newsletters that reference local events or consumer habits can draw a tighter connection with the consumer and boost loyalty.

The cultural context encompasses more than just language — it includes humor, idioms, symbols, and societal norms, all of which can vary dramatically from one region to another. Ignoring these elements when localizing content can lead to misunderstandings and offend audiences or simply result in content that fails to engage its intended audience.



REMEMBER Don't think only about your marketing content that appears in text. AI tools can assist with subtitling and captioning for video content, too, which can help ensure that your multimedia materials are accessible to a broader audience.

Harnessing multilingual large language models

A relatively recent development in the world of generative AI involves the development of multilingual large language models (LLMs). A multilingual LLM is an advanced AI system designed to understand, generate, and process text across multiple languages. It leverages vast amounts of text data from various languages to learn linguistic patterns, idioms, and cultural nuances, which enables it to perform a wide range of language-related tasks. For example:

- **For translation activities,** multilingual LLMs can seamlessly convert text from one language to another while maintaining the original meaning, tone, and context. This capability is crucial for businesses and organizations aiming to communicate effectively in a global market, ensuring that business messaging resonates with diverse audiences.
- **For localization activities,** these models go beyond mere translation, adapting content to align with specific cultural norms and preferences. Multilingual LLMs can localize marketing materials, websites, and software applications to make products and services more accessible and appealing to users worldwide.

Although these tools are still emerging and aren't perfectly primed for marketing use cases, you can still evaluate them in the context of your marketing translation and localization efforts. As with any AI tools, they also require human oversight. Table 6-1 lists LLMs that you may find useful.

TABLE 6-1 LLMs for Localization and Translation

Development Company	*LLM*	*What It Is and Does*
OpenAI	GPT-4	Understands and generates text in multiple languages, enabling diverse applications, including translation, content creation, and conversation
Google	mT5 (Multilingual Text-to-Text Transfer Transformer)	An extension of the T5 model, trained on a large multilingual data set, mT5 excels at translation, summarization, and question-answering tasks across languages
Meta	XLM-R (Cross-lingual Language Model —	Improves language understanding across many

Development Company	*LLM*	*What It Is and Does*
	Roberta)	languages, focusing on tasks such as text classification and sentiment analysis
Hugging Face	BERT (Bidirectional Encoder Representations from Transformers) Multilingual	Built on Google's BERT architecture, this model supports more than 100 languages for tasks such as named-entity recognition and question answering
Microsoft	Turing Multilingual Language Model	Used in applications such as translation, content moderation, and more efficient search algorithms

To use ChatGPT (GPT-4) for translation, follow these steps:

1. **Identify the text that you want to translate and indicate the source and target languages.**
   Keep in mind that ChatGPT's capabilities for translation may vary from language to language.
2. **Input the text into ChatGPT and clearly mention the translation request once more by sharing which language you want to translate it to.**
   ChatGPT processes the text by using its understanding of multiple languages to generate a translation that appears immediately below your input.
3. **Review the translation for accuracy.**
   Keep in mind that ChatGPT is a powerful tool, but it may not always match the quality of a professional human translator, especially for complex or nuanced texts.



REMEMBER If you're a small business marketer, accessing the other multilingual LLMs listed in Table 6-1 is a little more complex. And as with ChatGPT, reviewing the results for accuracy is critical. The models are reliant on the data used to train them, and the data can result in translations that include bias and cultural insensitivity.

Applying AI's capabilities

Advancements in AI technology enable content creators to produce a variety of localized versions of their work more efficiently and to provide a digital experience that feels deeply personalized. For example, you can customize AI-generated visual content to reflect cultural symbols, motifs, and color schemes that appeal to specific audiences.



REMEMBER Customer expectations have increased dramatically over the last few years, as evidenced by a 2023 Salesforce article titled "What Are Customer Expectations, and How Have They Changed?" — not to spoil it, but subtitled "Customer expectations hit all-time highs." (To access this article, go to `www.salesforce.com` and enter "customer expectations" in the search text box. The article should pop up in the results.) Customers expect brands to understand them and talk to them on their terms in ways that they appreciate and relate to. Using AI to effectively translate and localize all forms of content makes the effort required to meet heightened customer expectations much cheaper and quicker to do.

AI tools offer sophisticated solutions that can help your business navigate the cultural complexities and linguistic challenges of localizing and translating your customer-facing content. You can take advantage of these capabilities of AI tools and language models to fine-tune your websites, advertising, and online marketing:

- **Broad, inclusive data analysis:** AI tools can analyze large data sets to identify cultural trends, preferences, and sensitivities specific to different audience segments. For example, AI-powered content analysis tools can scrutinize social me-

dia behavior, search queries, and online discussions across various cultures to glean insights into what content resonates in which regions. Because cultural trends and preferences can shift quickly, AI tools that continuously analyze online communications and train on real-time data are the most valuable.

- **Content adaptation for local relevance:** AI can assist in the localization process by identifying and adapting content that must be customized for cultural relevance. These adaptations may include modifying color schemes on a website to match cultural preferences or adjusting the tone and style of a customer relationship management (CRM) newsletter to align with local communication norms.
- **Language adjustment to retain content essence:** Certain AI models can detect and adapt to subtle linguistic and cultural nuances, a capability that helps ensure that translations maintain the original message's intent, humor, and emotional tone. For instance, if you need to develop copy for a multinational campaign, choose AI tools that can suggest language adjustments to avoid cultural taboos and embrace local expressions. When you use these tools to translate, they don't literally translate the text word-for-word, but ideally include the local flavor — idioms, for example — that keeps the essence of the communication whole.

TRANSLATION GONE WRONG

In 2014, KFC's entry into the Chinese market hit a snag when their iconic slogan "Finger Lickin' Good" was incorrectly translated as "Eat Your Fingers Off." This error led to widespread amusement and bewilderment among Chinese consumers, sparking a flurry of social media activity that highlighted the mistranslation. The incident underscored the challenges of accurately conveying brand messages across languages and cultures, and it emphasized the importance of meticulous localization to avoid misinterpretations that can alienate potential customers. The translation was literally correct but localized incorrectly. Human oversight is probably the only way that this mistranslation could have been avoided. In this instance, AI tools may not have picked it up because the phrase being translated was an idiom specific to the English language.



REMEMBER Although scores of AI tools can help you with translation, localization, personalization, and sentiment analysis for all your customer-facing content, having a human eye to look over what the AI engines generate is essential. Sometimes, AI engines get it wrong (less so now than before, but it still happens). You don't want to risk damage to your brand or your relationship with your customers by an AI-generated bad translation or a missed cultural nuance.



TIP Engage AI tools to help personalize content at scale — for the entire breadth of your business's reach — and enable your marketers to create variations of a campaign that speak directly to the cultural context of each segment of their global audience. AI-powered *transcreation* (a term for adapting content creatively and culturally by using AI to maintain its original intent and emotional impact across languages and cultures) can help create culturally relevant and creative translations of your brand messaging that resonates with local audiences.

Checking out AI tools you can use

In this section, I introduce some specific AI tools that you can find to assist with localization and translation. Tables 6-2 and 6-3 list tools for consumer and enterprise use, respectively. Neither table offers an exhaustive list, and many other AI localization and translation tools exist, each with its own strengths and weaknesses. Research and compare various tools to find the best fit for your specific needs.

TABLE 6-2 Consumer AI Tools for Translation

Tool	*Website*	*Features*
DeepL	`www.deepl.com`	A neural machine tool focused on European languages (see Chapter 5 for

Tool	*Website*	*Features*
		more information on neural networks)
Google Translate	`http://translate.google.com`	A free tool that supports more than 100 languages and can translate text, speech, and web pages
Microsoft Translator	`http://translator.microsoft.com`	A cloud-based translation platform that offers text and speech translation, as well as real-time conversation translation
Babelfish	`www.babelfish.com`	Uses human editors and AI to provide translations in more than 75 languages
iTranslate	`https://itranslate.com`	An app that supports more than 40 languages and offers text, voice, and offline translation capabilities
Phrase	`https://phrase.com`	Localization and translation platform featuring automation, AI customization, workflow orchestration, and analytics

Tool	*Website*	*Features*
Wordbee	`www.wordbee.com`	A management platform that offers AI-powered translation, localization, and project management tools, as well as a community of translators

TABLE 6-3 Enterprise AI Tools for Translation

Tool	*Website*	*Features*
Acrolinx	`www.acrolinx.com`	A content creation and translation management platform that works across multiple languages and channels
RWS Tridion	`www.rws.com/content-management/tridion`	A content management and translation management platform that offers AI-powered translation, localization, and

Tool	*Website*	*Features*
		project management tools
Sitecore	`www.sitecore.com`	A customer experience platform that offers AI-powered translation and localization capabilities, as well as tools for managing and delivering personalized customer experiences
Drupal	`https://new.drupal.org/home`	An open-source content management platform that offers AI-powered translation and localization capabilities through various modules

Tool	*Website*	*Features* and integrations
WordPress	`https://wordpress.org`	A content management platform that offers various AI-powered translation and localization plugins, such as WPML and Polylang

Tool	*Website*	*Features*
HubSpot	`www.hubspot.com`	A marketing, sales, and customer-service platform that offers AI-powered translation and localization capabilities, as well as tools for managing and delivering personalized customer experiences
Salesforce	`www.salesforce.com`	A customer relationship management platform that offers AI-powered translation and localization capabilities, as well as tools for managing and deliver-

Tool	*Website*	*Features*
		ing personalized customer experiences
IBM Watsonx	www.ibm.com/watsonx	An AI platform that offers various language translation and localization tools, including machine translation, natural language understanding, and natural language generation through its Watson Language Translator

Tool	*Website*	*Features*
Lilt	`http://lilt.com`	A platform that uses AI and human editors to provide translations in more than 50 languages
Smartcat	`www.smartcat.com`	A management platform that offers translation, localization, and project management tools
Google Cloud Translation	`http://cloud.google.com/translation`	A cloud-based translation platform that offers AI-powered translation and localization capabilities, as well as tools for managing and delivering person-

Tool	*Website*	*Features*
		alized customer experiences

Adopting Core Strategies for Localization

To harness AI effectively for localization and translation, you can adopt several core strategies that ensure your digital content is both accurately translated and also culturally relevant and engaging. Some of these strategies may be more appropriate for larger organizations, but understanding them can help you determine your own language translation strategies, even if you're marketing for a smaller business. Increasingly, you can find companies that cater to small and medium businesses, offering to assist in building functionality to support localization and translation strategies.

Leveraging machine learning

You can leverage machine learning for contextual translation by using these tools and practices:

- **Training AI models by using domain-specific data:** Customize machine learning models by using industry-specific data sets to improve the accuracy of translations. These data sets can help ensure that your efforts correctly translate technical and professional jargon. You can also use *natural language processing* (NLP), which analyzes and interprets natural language, allowing for more accurate translations and localizations.

  

  REMEMBER The quality of the data used to train an AI tool directly impacts the quality of the translations it produces. Ensure that the training data is high-quality, relevant, and aligned with your brand's tone and style.
- **Using neural machine translation (NMT):** Employ *NMT,* which uses neural networking to predict the possibility of a sequence of words, to understand and translate entire sentences or paragraphs at the same time, rather than word-by-word. NMT tools can capture context (see the section "Capturing cultural con-

text," earlier in this chapter) and reduce errors, thus providing more fluid and natural translations.

- **Implementing post-editing processes:** Use AI to flag translations that may require human review. Combining the efficiency of AI with the nuanced understanding of human translators helps you refine content for local markets.

Adopting AI-driven cultural adaptation tools

Check out AI algorithms, tools, and techniques that can help you customize and personalize your content for the culture of the customers you want to reach:

- **Analyze cultural preferences with AI.** Analyze local consumer behavior and preferences, allowing for the adaptation of content, images, and design elements to match local tastes.
- **Deploy sentiment analysis.** Gauge the emotional tone of content and adjust messaging so that it's culturally appropriate and it resonates with the target audience's values and expectations.
- **Customize content for regional trends.** Dynamically update content based on trending topics, local news, and cultural events, ensuring that the brand remains relevant and engaged with local communities.

Enhancing personalization and localization efficiency

AI tools and systems, by design, can add efficiency to workflows. In the case of localization and translation, the following list highlights areas where AI can strategically streamline processes and learn in real time:

- **Automating content localization workflows:** Streamline the localization process. Specifically, AI can identify content that needs translation and also assign tasks and manage workflows to accomplish the translation and reduce time-to-market for localized content.



TIP Consider implementing a *translation memory system,* which stores previously translated content in a database. You can access this database to retrieve and reuse previously translated content — a practice that can reduce costs and improve efficiency.

- **Personalizing user experiences at scale:** Tailor digital experiences, including website content, social media, and e-mail marketing. AI can learn to recognize

individual user preferences and local cultural norms so that it can enhance content aimed at increasing user engagement and satisfaction.

- **Continuous learning and improvement:** Learn from user interactions and feedback to continuously improve the quality of translations and the effectiveness of localized content. You can use machine learning (ML) algorithms that train by using large data sets of previously translated content to identify patterns and relationships between languages. Doing so can help ensure that digital experiences remain relevant and compelling over time. In addition, these ML practices can help improve the accuracy of translations and reduce the need for human intervention.

Controlling quality when using AI

Using AI for quality control in localization and translation can help improve accuracy, efficiency, and consistency in your marketing content and campaigns. Here are some tips and tricks for teaming up with AI effectively in your quality-control activities:

- **Use human post-translation editing.** Although AI-powered translation and localization efforts are highly accurate, you may want to add a human post-translation editing phase to refine the AI's output and ensure it meets the desired level of quality that you want for your localization efforts.



TIP AI tools can be helpful and efficient, but a human editor should review and polish translations, especially for critical content such as branding materials, taglines, and mission statements.

- **Integrate your localization and translation tools with other marketing tools.** Real-time localization and translation solutions often integrate with popular marketing tools, such as content management systems (CMSs), customer relationship management (CRM) software, and marketing automation platforms. Making sure that these systems work together adds fluidity and consistency to your marketing content and messages.
- **Use a hybrid approach.** When I talk about a *hybrid approach* for your localization and translation, I mean combining human translation with machine learning algorithms. Adding human knowledge and experience with languages and cultures to your AI-driven translation efforts can help ensure that the translation is accurate and culturally relevant.



TIP Combine the strengths of both human translators and AI tools to achieve the best results. AI tools can handle large volumes of content quickly and accurately, and human translators can review and refine the translations to ensure that they meet your brand's standards.

- **Use a glossary and a style guide.** Basic reference materials — such as a glossary and style guide — can infuse localization and translation output with an extra polished feel. You can create a glossary of technical terms and jargon, as well as a style guide (which specifies formatting, tone, and other style issues), which AI and humans can reference during the translation and review processes. When you share these references with all team members involved in the process, you can help to ensure consistency in translation output.
- **Monitor and analyze.** Continuously monitor and analyze the performance of the AI-powered localization and translation solutions that you use to identify areas for improvement. If you find that the solution is not meeting your needs, either determine the settings you can adjust to increase the accuracy of your translations or change the solutions completely.

Examining Real-Time Localization and Translation Solutions

Real-time localization and translation solutions use AI to translate and localize content in real time. This speedy processing allows businesses to reach new markets and engage with customers of varying languages and cultures immediately when those customers have questions. In the following sections, you can get a look at how real-time localization and translation solutions work, their benefits, and their applications in marketing.

Seeing how real-time solutions work

Real-time localization and translation solutions use AI algorithms that train on vast amounts of data, which include text in multiple languages and content that reveals cultural nuances. For example, the idiom "break a leg" in English would translate into German literally, implying a harmful wish. This type of training enables the AI to learn patterns and relationships between languages so that it generates more accurate translations and localizations.

The foundation for and process of real-time localization and translation involves several phases:

- **Content analysis:** AI algorithms analyze the source content and identify the most critical elements that require localization or translation. This process, known as *content segmentation,* ensures that the AI translates or localizes only the parts of the content that are required to convey the primary message from the original language to the relevant target audience.
- **Translation memory:** Memory-based translation, also known as *translation memory,* is a technique that stores previously translated content in a database. When similar content is encountered again, the AI can draw upon this database to produce consistent, high-quality translations, which reduces the need for human intervention. The section "Enhancing personalization and localization efficiency," earlier in this chapter, discusses the use of translation memory.
- **Real-time translation:** The trained AI models, along with natural language processing (NLP) algorithms, enable real-time translation of text, speech, or other forms of communication. In marketing, your business can take advantage of the real-time aspects of AI processing to quickly adapt its content for various regions and languages, ensuring that your messaging reaches a global audience.
- **Localization:** Localization goes beyond translation, encompassing not only language but also cultural adaptations, formatting, and content optimization for a specific region. AI-powered localization solutions analyze the source content and adapt it to the target culture, considering local preferences, norms, and regulations.

Recognizing the benefits of real-time solutions

Real-time localization and translation solutions enable businesses to reach new markets and engage with their audience in various languages and cultures, with the goal of improving their customers' experience and driving growth and revenue. The benefits of using these solutions are numerous, including

- **Cost savings:** Reduce or limit the need for human intervention, saving businesses time and money.
- **Increased efficiency:** Reach new markets and engage with a new customer base quickly.
- **Improved accuracy:** AI algorithms that train on vast data resources can help ensure that translations are accurate and culturally relevant. This increased accuracy reduces the risk of misunderstandings or miscommunications. When the

solutions use translation memory (see the preceding section), the accuracy is further improved.

- **Enhanced customer experience:** Provide your customers with a more personalized and engaging experience. The goal is to increase customer satisfaction and loyalty.

Applying real-time solutions in marketing

Real-time localization and translation solutions can help your business efficiently and effectively adapt your marketing strategies for a global audience. Here are some of the solutions' numerous applications in marketing:

- **Website localization:** Localize your business's websites for various markets and customer bases. This localization helps ensure that your online marketing presence is felt globally. And don't forget about search engine optimization (SEO); optimize your translated content for local search engines by using local keywords and phrases to improve visibility and drive organic traffic.

  

  REMEMBER For websites especially, collaborate with local teams and in-country experts to ensure that your translations are accurate, culturally relevant, and aligned with local preferences.
- **Content marketing:** Because you can create content in one language and translate it into multiple languages by using real-time localization and translation solutions, you can more quickly produce content that resonates with your target audience, regardless of their language or location.
- **Social media marketing:** Engage with your audience on platforms other than your business's own websites or shopping sites.
- **E-mail marketing:** Connect with your audience on a direct, personal level, regardless of their language or location.
- **Advertising:** Reach your target audience in different markets and languages.